Exact dynamics of Eratosthenes sieve beyond $2p$: coincident fusions, births of gaps and missing differences

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Abstract

At each stage of Eratosthenes sieve the integers coprime to a modulus N form a periodic cycle of gaps $\mathcal{G}(N)$. F. B. Holt showed that the populations of gaps and constellations, together with their driving terms, evolve by one linear system, the same for all of them, as long as the span is smaller than twice the next prime q, and that beyond this threshold several points of one driving term are removed in the same image exactly when q divides their distance. Part I gives the exact transition law for all spans: the points of a driving term split into residue classes modulo q, each class is removed in exactly one of the q images, and this determines every image. Consequences are a form in binomial moments with nonnegative sources, an exact birth law, a criterion for first appearance, and a necessary and sufficient condition for the exactness of Holt's Markov chain on lengths. The transition law, the birth law and the criterion for Holt's chain were verified exhaustively for every constellation of span at most 40 on all cycles up to 19#. As applications we extend Holt's exact models to gaps $g \le 150$ and list all their coefficients, classify the missing differences of Ziller's table up to $k = 30$, describe how the maxima of $\mathcal{G}(p_n\#)$ arise for $n \le 64$, and test Holt's survival hypothesis on all primes between 47^2 and 10^9: the gaps, and the pairs and triples of consecutive gaps, between primes near x follow the cycle at the level x^{1/e^γ} of the Banks–Ford–Tao model within sampling noise, and not at the level $\sqrt{x}$. Part II concerns windows with the least number of survivors, the function $\varphi_{\min}$ of Costello and Watts. The minimum for qN is attained by windows for N close to the minimum: we prove that the excess, the depth, is 0 for spans below $6q$ and at most 1 below $8q$; in all 3943 computed cases with spans up to about $20q$ it is at most 1, and in all computed windows up to about $230q$ at most 2. A certified recursion computes $\varphi_{\min}$ from the previous stage; in our tests it was faster than direct search.

1 Introduction

For a prime p let $p\#$ denote the product of all primes up to p. The integers coprime to $p\#$ — the candidates that survive Eratosthenes sieve up to p — form a periodic set; the sequence of differences between consecutive elements over one period is the *cycle of gaps* $\mathcal{G}(p\#)$, of length $\varphi(p\#)$ and span $p\#$. Holt and Rudd [1] observed that $\mathcal{G}(p_{k+1}\#)$ arises from $\mathcal{G}(p_k\#)$ by a simple recursion: concatenate p_{k+1} copies of the cycle and fuse adjacent gaps at the positions of the multiples of p_{k+1}. In a series of papers [1, 2, 3, 4] Holt developed this into a discrete dynamical system. A central notion is that of a *driving term*: a run of consecutive gaps whose sum equals a given gap g (or which contains a given constellation s), and which can turn into g (or into s) at a later stage by fusions. The populations $n_{g,j}(p\#)$ of driving terms of span g and length j obey the linear recursion

$$n_{g,j}(qN) = (q - j - 1)\, n_{g,j}(N) + j\, n_{g,j+1}(N)$$

with explicit eigenstructure, which yields exact models for the relative populations of gaps and constellations at all later stages of the sieve.

This recursion holds under a restriction: the span must be smaller than twice the next prime. For larger spans two fusions can fall into the same image of a driving term, and Holt characterized when this happens: two fusions fall into the same image if and only if the next prime divides their distance [3, Lemma 3.1]. Holt notes that the minimum distance between fusions is $2p_{k+1}$ [4, Obs. 2.3], that for spans below $2p_1$ all fusions of a constellation fall into different images [4, Lemma 2.4], and that for gaps beyond $2p_1$ the counts of driving terms of various lengths could not be determined [1, 2]; the single case $g = 2p_1$ was handled by a special first iteration [2]. Several fusions at the *boundary* of one image are accounted for in his aggregate count through the number $\nu_s(p)$ of residue classes covered by the boundary points [4, Lemma 2.5, Lemma 2.7]. What remained open is the case of several *interior* fusions in the same image, which changes the distribution of lengths. For single windows, the removal of one residue class per future prime appears in Samuel's residue slices [19]. His birth-delay identity [19, Thm. 4.2] expresses the delay before a gap first appears as a minimum over single windows, while Corollary 4.3 is a one-step criterion in terms of populations; his collision profiles use binomial moments of the class occupancies of one window [19, Prop. 5.1], while Theorem 3.7 concerns binomial moments of populations; Tucker's atlas [20] stratifies coverings of primorial windows by their defect.

Main result. We show that the transition from $\mathcal{G}(N)$ to $\mathcal{G}(qN)$ is determined exactly, for every span, by the partition of the points of a driving term into residue classes modulo q: each class is removed in exactly one of the q images, and the remaining images are untouched (Theorem 3.4). The partition into classes is Holt's criterion [3, Lemma 3.1] applied to all points of a driving term, interior points included; what is new is that this partition determines the lengths of all images, together with the consequences drawn from it. Holt's recursion is the special case in which all classes are singletons.

Consequences. Section 3 derives a ladder of thresholds for the size of the classes (Corollary 3.6) and a form of the dynamics in binomial moments, in which the non-uniformity appears only as a nonnegative source (Theorem 3.7). Section 4 gives an exact birth law for gaps and constellations (Theorem 4.1), a criterion for the first appearance of a constellation (Corollary 4.3), and the description of the set of gap lengths $\mathcal{D}(qN)$ in terms of $\mathcal{G}(N)$. Section 5 identifies the exact process on shapes of driving terms and shows that Holt's Markov chain on lengths is exact if and only if no populated shape has two points congruent modulo q (Theorems 5.1 and 5.2).

Applications. We extend Holt's exact models from gaps $g \leq 82$ to $g \leq 150$, which includes the gaps $84 \leq g \leq 90$ named in [4] as a current goal; reproduce the rows of Holt's worked example [4, Fig. 6] that lie outside the uniform regime; classify all missing differences in Ziller's table [5] up to $k = 30$; and describe how the maximal gaps of $\mathcal{G}(p_n\#)$ — the values of Jacobsthal's function for primorials, OEIS A048670 — arise from the previous cycle for $n \leq 64$; and we test Holt's survival hypothesis [16] on all primes between 47^2 and 10^9 (Section 6.4).

Part II. Sections 8 and 9 study windows of fixed length with the least number of points of $\Omega(N)$ and the depth, above this minimum, of the windows from which the minimum for qN arises.

What is not claimed. All results concern the cycles $\mathcal{G}(N)$; nothing is proved about the actual gaps between primes; Section 6.4 is an empirical comparison. Ziller's Conjecture 4.1 [5] is reformulated, not proved. In Part II the bound on the depth is proved only for spans

below $8q$; beyond, the depth is computed. Statements marked as computed or observed rest on computation and are labelled as such.

2 Setting and notation

Throughout, N is a squarefree integer with $2 \mid N$, and q is an odd prime with $q \nmid N$. The main case is $N = p_k\#$ and $q = p_{k+1}$. Let $\Omega(N) = \{x \in \mathbb{Z} : \gcd(x, N) = 1\}$, a set periodic modulo N; for $N = p\#$ these are the p-rough numbers. The cycle of gaps $\mathcal{G}(N)$ is the sequence of differences of consecutive elements of $\Omega(N)$ over one period; $\mathcal{D}(N)$ is the set of gap lengths occurring in $\mathcal{G}(N)$ and

$$H(N) := \max \mathcal{D}(N)$$

is the largest gap, the value of Jacobsthal's function at N. For primorials we write $\mathcal{D}(k) := \mathcal{D}(p_k\#)$ and $h(k) := H(p_k\#)$, as in [5].

Definition 2.1 (Constellations and driving terms)**.** A *constellation* is a finite sequence $s = (s_1, \ldots, s_m)$ of even positive integers with span $|s| = s_1 + \cdots + s_m$; its points are $C = \{c_0 = 0 < c_1 < \cdots < c_m = |s|\}$, $c_i = s_1 + \cdots + s_i$. We assume $0 < |s| < N$. A *driving term for* s in $\mathcal{G}(N)$ is a residue $x \bmod N$ with $x + c \in \Omega(N)$ for all $c \in C$. Its point set is

$$P(x) = \{t \in \{0, 1, \ldots, |s|\} : x + t \in \Omega(N)\} \supseteq C,$$

its *length* is $J = |P(x)| - 1$ (the number of gaps of $\mathcal{G}(N)$ it spans), and $E(x) = P(x) \setminus C$ is the set of its *extra points*. We write $n_{s,J}(N)$ for the number of driving terms for s of length J; the occurrences of s itself are those with $J = m$. For a single gap ($m = 1$, $s = (g)$) we write $n_{g,j}(N)$.

The condition $|s| < N$ is a convention ensuring that a driving term lies within one period of the cycle; none of the proofs below uses it (Remark 4.5). In Holt's terms, a driving term for s becomes s through *interior fusions* (removal of extra points), while a *boundary fusion* (removal of a point of C) eliminates it as a driving term for s [4, §2.1].

Images and classes. For $x \bmod N$ the q residues $X \bmod qN$ with $X \equiv x \pmod N$ are $X_i = x + iN$, $0 \le i < q$; we call them the *images* of x. For a driving term x we partition $P(x)$ into classes of congruence modulo q and denote the set of classes by $\mathcal{C}(x)$. For a finite set A, $\nu_q(A)$ denotes the number of residue classes modulo q occupied by A; thus $|\mathcal{C}(x)| = \nu_q(P(x))$, and $\nu_q(C)$ is Holt's $\nu_s(q)$.

3 The transition law

Lemma 3.1. *For $t, t' \in P(x)$ we have $t \equiv t' \pmod q$ if and only if $2q \mid t - t'$.*

Proof. Since $2 \mid N$, all elements of $\Omega(N)$ are odd, so $t - t' = (x + t) - (x + t')$ is even; for even d and odd prime q, $q \mid d \iff 2q \mid d$. □

This is the quantitative content of Holt's observation that the minimum distance between fusions is $2p_{k+1}$ [4, Obs. 2.3].

Lemma 3.2. *The images $X_0, \ldots, X_{q-1}$ of x run through all residues modulo q, each exactly once.*

Proof. $\gcd(N, q) = 1$, so $i \mapsto iN \bmod q$ is a bijection of $\mathbb{Z}/q$. □

Lemma 3.2 is [3, Lemma 4.1], [4, Lemma 2.6]. It gives Holt's criterion [3, Lemma 3.1]: two points of a driving term are removed in the same image if and only if q divides their distance, or equivalently, by Lemma 3.1, $2q$ divides it. Holt applies it to the boundary points, to count admissible instances; Theorem 3.4 applies it to all points.

Lemma 3.3. *If $X \equiv x \pmod N$, then $X+t \in \Omega(qN)$ if and only if $x+t \in \Omega(N)$ and $q \nmid X+t$.*

Proof. $\gcd(X+t, qN) = 1$ iff $\gcd(X+t, N) = 1$ and $q \nmid X+t$, and $X+t \equiv x+t \pmod N$. □

Theorem 3.4 (Transition law)**.** *Let x be a driving term for s of length J in $\mathcal{G}(N)$, with classes $\mathcal{C} = \mathcal{C}(x)$. Among the q images of x:*

1. *for each class $K \in \mathcal{C}$ there is exactly one image X_K in which $q \mid X_K + t$ for all $t \in K$ and for no $t \in P(x) \setminus K$;*
2. *the remaining $q - |\mathcal{C}|$ images contain no point divisible by q.*

Consequently the images of type 2 are driving terms for s of length J in $\mathcal{G}(qN)$; the image X_K with $K \cap C = \emptyset$ is a driving term for s of length $J - |K|$; and the image X_K with $K \cap C \neq \emptyset$ is not a driving term for s. Every driving term for s in $\mathcal{G}(qN)$ arises in this way from exactly one driving term in $\mathcal{G}(N)$ and exactly one of its images. Hence

$$n_{s,J'}(qN) = \sum_x \Big([\, J(x) = J' \,] \big(q - |\mathcal{C}(x)|\big) + \#\{K \in \mathcal{C}(x) : K \cap C = \emptyset,\ J(x) - |K| = J'\} \Big), \quad (1)$$

the sum running over all driving terms for s in $\mathcal{G}(N)$.

Proof. $q \mid X + t$ iff $X \equiv -t \pmod q$. The set $\{-t \bmod q : t \in P(x)\}$ has exactly $|\mathcal{C}|$ elements, one for each class. By Lemma 3.2 each residue modulo q is taken by exactly one image; the image with residue $-t$ removes exactly the points of the class of t, and the other $q - |\mathcal{C}|$ images remove none. By Lemma 3.3 the points of $\Omega(qN)$ in $[X, X + |s|]$ are the points of $P(x)$ minus the removed class. If the removed class meets C, a point of s is lost; otherwise $|K|$ extra points are removed and the length drops by $|K|$. Conversely, if X is a driving term for s in $\mathcal{G}(qN)$, then by Lemma 3.3 $x = X \bmod N$ has $x + c \in \Omega(N)$ for all $c \in C$, so x is a driving term in $\mathcal{G}(N)$ and X is one of its images; x is determined by X. □

Remark 3.5 (Holt's recursion). If no two points of $P(x)$ are congruent modulo q — by Lemma 3.1 this holds whenever $|s| < 2q$ — all classes are singletons, $|\mathcal{C}(x)| = J + 1$, and there are $J - m$ extra points. Then (1) becomes

$$n_{s,J}(qN) = (q - J - 1)\, n_{s,J}(N) + (J + 1 - m)\, n_{s,J+1}(N),$$

which for gaps ($m = 1$) is the recursion of Holt and Rudd [1, Thm. 3.2], and for constellations the recursion underlying [4, Fig. 7]. Theorem 3.4 determines the lengths also when several interior fusions fall into one image, the case left open in [1, 2].

Corollary 3.6 (Ladder of thresholds)**.** *A class with $r \geq 2$ points has its extreme points at distance at least $2q(r-1)$, a positive multiple of $2q$. If moreover $3 \mid N$ and $q \neq 3$, a class with three points spans at least $6q$.*

Proof. The first statement follows from Lemma 3.1. For the second, let $t_1 < t_2 < t_3$ lie in one class, $t_2 = t_1 + 2qa$, $t_3 = t_2 + 2qb$ with $a, b \geq 1$. If $a = b = 1$, the residues of $x + t_1, x + t_1 + 2q, x + t_1 + 4q$ modulo 3 are $x + t_1, x + t_1 - q, x + t_1 + q$, all three residues since $3 \nmid q$; one of these numbers would be divisible by 3, contradicting $3 \mid N$. Hence $a + b \geq 3$. □

The second statement is the case of three points of Hagedorn's bound for the number of positions a prime can remove [6, Prop. 3.3], which he attributes to Haugland. The same argument gives further thresholds. If $3 \mid N$ and $q \neq 3$, write the points of a class as $t_1 + 2qa_i$; since the numbers $x + t_1 + 2qa_i$ are prime to 3, the integers a_i avoid one residue class modulo 3. Four integers of diameter at most 3, and five of diameter at most 5, meet every residue class modulo 3, so classes of four points span at least $8q$ and classes of five points at least $12q$. In all our tests (Section 7) the first classes of $2, 3, 4, 5$ points occur exactly at the spans $2q, 6q, 8q, 12q$. In particular, for $3 \mid N$ and $|s| < 6q$ all classes have at most two points, at distance $2q$ or $4q$.

Theorem 3.7 (Moment form)**.** *For $k \geq 0$ let $S_k(s, N) = \sum_J \binom{J-m}{k} n_{s,J}(N)$, the number of pairs (driving term, k-subset of its extra points), and for a finite set A let $V_N(A) = \#\{x \bmod N : x + a \in \Omega(N)\ \forall a \in A\}$. Then*

$$S_k(s, N) = \sum_{\substack{T \subseteq \{1,\ldots,|s|-1\} \setminus C \\ |T|=k}} V_N(C \cup T), \qquad S_k(s, qN) = (q - m - 1 - k)\, S_k(s, N) + \Pi_k(s, N),$$

where

$$\Pi_k(s, N) = \sum_{|T|=k} \big(m + 1 + k - \nu_q(C \cup T)\big)\, V_N(C \cup T) \ \geq\ 0.$$

Moreover $n_{s,m}(N) = \sum_{k \geq 0} (-1)^k S_k(s, N)$.

Proof. A pair (x, T) with $x + a \in \Omega(N)$ for $a \in C \cup T$ is exactly a driving term x together with a k-subset T of its extra points, which gives the first identity. By the Chinese remainder theorem $W_{qN}(A) = V_N(A)\,(q - \nu_q(A))$, since the condition modulo q excludes exactly $\nu_q(A)$ residues. Writing $q - \nu_q(A) = (q - |A|) + (|A| - \nu_q(A))$ with $|A| = m + 1 + k$ gives the recursion; $|A| - \nu_q(A) \geq 0$ counts coincidences modulo q. The last identity is binomial inversion, $\sum_k (-1)^k \binom{J-m}{k} = [J = m]$. □

For $k = 0$ the theorem gives $S_0(s, qN) = (q - \nu_q(C))\, S_0(s, N)$, the aggregate count of Holt [3, Cor. 1.2], [4, Thm. 1.1, Lemma 2.7] (for gaps [1, Cor. 5.3]). The coefficients $q - m - 1 - k$ are the eigenvalues of Holt's uniform system and the S_k are its left-eigenvector coordinates [4, §3]; in these coordinates the non-uniformity is a nonnegative source, and decreases of a population can arise only through the alternating sum for $n_{s,m}$.

4 Births

Theorem 4.1 (Birth law)**.**

$$n_{s,m}(qN) = \big(q - \nu_q(C)\big)\, n_{s,m}(N) + B_s^{(q)}(N),$$

where $B_s^{(q)}(N)$ is the number of driving terms for s in $\mathcal{G}(N)$ whose extra points form a single residue class modulo q that is disjoint from the classes of the points of C.

Proof. Apply Theorem 3.4 with target length m. Occurrences of s in $\mathcal{G}(N)$ have $P = C$ and keep $q - \nu_q(C)$ untouched images. A longer driving term yields an occurrence of s exactly when the removed class equals its whole set of extra points and contains no point of C, and then it yields exactly one. □

For a gap ($s = (g)$) we write $e_g = [\,2q \mid g\,]$, so that $q - \nu_q(\{0, g\}) = q - 2 + e_g$. In the language of constellations of consecutive gaps $(a_1, \ldots, a_j)$, $j \geq 2$, of $\mathcal{G}(N)$ with sum g, the condition of Theorem 4.1 reads: $2q \nmid a_1$, $2q \nmid a_j$, and $2q \mid a_i$ for $1 < i < j$ (Lemma 3.1). For $3 \mid N$ and $g < 6q$ only pairs (a, b) and triples (u, v, w) with $v \in \{2q, 4q\}$ occur. A pair with $2q \mid a$ or $2q \mid b$

does not produce a gap: we call this a *blocked birth*. A triple whose middle gap is a multiple of $2q$ and whose end gaps are not produces a gap of length $u+v+w$ in one image: a *coincident birth*.

Lemma 4.2 (Persistence). *If $\nu_q(C) < q$, then $n_{s,m}(qN) \geq n_{s,m}(N)$. If $\nu_q(C) = q$, then s occurs in no $\mathcal{G}(M)$ with $q \mid M$.*

Proof. The first part follows from Theorem 4.1; the second because every x then has a point $x + c$, $c \in C$, divisible by q. □

Holt's Theorem 1.1 [3] shows that every admissible instance of s, driving terms included, arises and persists; the lemma is the corresponding statement for the occurrences of s itself, in substance Holt's Lemma 2.6 [4].

Corollary 4.3 (First appearance). *Let $N_0 \mid N_1 \mid \cdots$ with $2 \mid N_0$ and $N_{i+1} = q_i N_i$ for odd primes $q_i \nmid N_i$. The constellation s occurs for the first time in this chain at N_{i+1} if and only if $n_{s,m}(N_i) = 0$ and $B_s^{(q_i)}(N_i) > 0$.*

Proof. If s occurs in $\mathcal{G}(N_{i+1})$ it is admissible modulo every prime dividing N_{i+1}; had it occurred at some N_r with $r < i$, Lemma 4.2 applied to $q_r, \ldots, q_{i-1}$ would give an occurrence in $\mathcal{G}(N_i)$. The converse follows from Theorem 4.1. □

Corollary 4.4 (Calm regime). *If all gaps of $\mathcal{G}(N)$ are smaller than $2q$, then for every even g with $2 \leq g < N$*

$$n_{g,1}(qN) = (q - 2 + e_g)\, n_{g,1}(N) + n_{g,2}(N),$$

$$\mathcal{D}(qN) = \mathcal{D}(N) \cup \{a + b : (a, b) \text{ adjacent gaps of } \mathcal{G}(N)\}, \qquad H(qN) = \max\{a + b\}.$$

Proof. No gap is a multiple of $2q$, so every pair qualifies and no constellation with a middle gap divisible by $2q$ exists. $\mathcal{D}(N) \subseteq \mathcal{D}(qN)$ because $q - 2 + e_g \geq 1$. □

Remark 4.5 (Domain). The corollaries concern gaps up to $H(qN)$ while the convention of Section 2 requires spans below N. This is consistent: $H(qN) \leq N$ for every even squarefree N with an odd prime factor. Indeed, by Kanold's bound $H(M) \leq 2^{\omega(M)}$ [8]; if $\omega(N) \geq 3$ then $2^{\omega(N)+1} \leq N$; if $N = 2p$ with $p \geq 5$ then $H(qN) \leq 8 \leq 2p$; and for $N = 6$ the gaps of $\mathcal{G}(6)$ alternate $4, 2$ and a prime $q \geq 5$ cannot remove two adjacent points, so $H(6q) = 6$. (For $N = 2$ one has $H(2q) = 4 > 2$.) The proofs of Theorems 3.4 and 4.1 do not use the convention at all, and our tests confirm the laws also for spans $\geq N$.

For primorials, the known values of Jacobsthal's function for primorials, $h(k) = H(p_k\#)$ (OEIS A048670; Holt writes $g_{\max}(p_k\#)$), give $h(k) < 2p_{k+1}$ for all $k \leq 18$ ($h(18) = 132 < 134$), while $h(19) = 152 \geq 142 = 2p_{20}$. Hence for $k \leq 18$

$$\mathcal{D}(k+1) = \mathcal{D}(k) \cup \{\text{sums of two adjacent gaps of } \mathcal{G}(p_k\#)\}, \qquad h(k+1) = \max\{\text{such sums}\},$$

and $k + 1 = 20$ is the first stage at which blocked and coincident births are possible. Ziller observed that $h(k) > 2p_k$ for $k > 17$ [5, §4]; the condition relevant here compares $h(k)$ with $2p_{k+1}$.

Corollary 4.6 (Missing differences). *An even g with $2 \leq g < H(qN)$ is missing from $\mathcal{D}(qN)$ if and only if $n_{g,1}(N) = 0$ and $B_g^{(q)}(N) = 0$. Let $M_2^{(q)}(N)$ be the largest sum $a+b$ over pairs of adjacent gaps of $\mathcal{G}(N)$ with $2q \nmid a$, $2q \nmid b$. A missing difference g is*

- *by cause: a* delay *if no pair of adjacent gaps of $\mathcal{G}(N)$ sums to g, and a* blocked birth *if such pairs exist but each contains a gap divisible by $2q$;*

- *by position: a* hole *if* $g < M_2^{(q)}(N)$*, and a* shadow *if* $M_2^{(q)}(N) < g < H(qN)$*; in the latter case* $H(qN) > M_2^{(q)}(N)$*, so* $H(qN)$ *is either* $H(N)$ *or arises from a coincident birth.*

In the calm regime every missing difference is a hole and a delay.

In all our data, whenever a shadow occurs, $H(qN)$ arises from a coincident birth (Table 2).

Corollary 4.7 (Maximal gaps)**.** *The largest gap* $H(qN)$ *is the maximum of* $H(N)$ *and of all spans* g *with* $B_g^{(q)}(N) > 0$*. If it is attained by a qualifying constellation of length* j*, then in each maximal interval of* $\mathcal{G}(qN)$ *produced by it the prime* q *alone removes exactly* $j - 1$ *points. For* $3 \mid N$ *and* $H(qN) < 6q$ *this is one point for a pair birth and two points, at distance* $2q$ *or* $4q$*, for a coincident birth.*

5 The process on shapes and Holt's chain

Let $m_P(N)$ be the number of driving terms for s with $P(x) = P$, the population of the *shape* P.

Theorem 5.1. *For every span the shape populations evolve by the exact linear system*

$$m_{P'}(qN) = \sum_P m_P(N)\, T_q(P, P'),$$

where $T_q(P, P) = q - |\mathcal{C}(P)|$*,* $T_q(P, P \setminus K) = 1$ *for every class* $K \in \mathcal{C}(P)$ *with* $K \cap C = \emptyset$*, and all other coefficients are zero. Holt's chain on lengths is the aggregation of shapes of equal length. If no populated shape has two points congruent modulo* q *— in particular if* $|s| < 2q$ *— the coefficients depend only on the length,* $T(J \to J) = q - J - 1$ *and* $T(J \to J - 1) = J - m$*, and Holt's chain is exact.*

The condition extends Holt's condition $|s| < 2p_1$ [3, §5], [4, Lemma 2.4], under which all fusions fall into separate images.

Proof. The system is Theorem 3.4. With singleton classes $|\mathcal{C}(P)| = J + 1$ and there are $J - m$ classes disjoint from C, each of one point. □

Theorem 5.2. *Holt's chain on lengths is exact at the transition from* N *to* qN *if and only if no populated shape has two points congruent modulo* q*. More precisely, let* J^* *be the largest length of a populated shape with two congruent points; then* $n_{s,J^*}(qN)$ *exceeds the value given by Holt's chain by*

$$\sum_{|P|=J^*+1} m_P(N)\,\big(|P| - \nu_q(P)\big) > 0,$$

the sum over the populated shapes of length J^**.*

Proof. Sufficiency is Theorem 5.1. Shapes longer than J^* have only singleton classes, so each image loses at most one point; such a shape contributes to length J^* only if its length is $J^* + 1$, and then through its $J^* + 1 - m$ extra points, as in Holt's chain. A shape P of length J^* keeps $T_q(P, P) = q - \nu_q(P)$ images of length J^*, which exceeds the coefficient $q - J^* - 1$ of the chain by $|P| - \nu_q(P)$, positive exactly when P has two congruent points. Shorter shapes do not contribute to length J^*. □

Following Holt's suggestion, we say that s is in the *uniform regime* at the transition from N to qN if no populated shape has two points congruent modulo q, and in the *non-uniform regime* otherwise. By Theorem 5.2 this is exactly when Holt's chain on lengths is exact; Holt's

condition $|s| < 2p_1$ is sufficient. All transitions are linear, by Theorem 5.1; what distinguishes the uniform regime is that the same matrix applies to all constellations of a given length.

If the only coincidences are among the points of s, then every populated shape has the same classes containing points of s and singleton classes otherwise, so by Theorem 5.1 a chain on lengths with a corrected diagonal is still exact, as for gaps divisible by $2q$.

6 Applications

6.1 Holt's exact models for gaps $84 \le g \le 150$

Holt's exact models require initial conditions $n_{g,j}(p_0\#)$ at a stage after which every step is uniform for the gap considered, that is $g < 2p_1$ for the prime p_1 following p_0. With initial conditions from $\mathcal{G}(37\#)$ this gives the models for $g \le 82$; the gaps $84 \le g \le 90$ are named in [4, §5] as a current goal.

We computed the complete matrices $n_{g,j}(p\#)$ for $p = 41$ ($g \le 110$), $p = 43$ ($g \le 104$) and $p = 47$ ($g \le 104$). Since $2 \cdot 53 = 106 > 104$, every step from $47\#$ on is uniform for $g \le 104$, so $n_{g,j}(47\#)$ are exact initial conditions, with $p_0 = 47$, for all gaps up to 104. They pass three independent checks.

1. The transitions $37\# \to 41\# \to 43\# \to 47\#$, all of which contain spans beyond $2q$, are reproduced exactly by Theorem 3.4 from Holt's data for $\mathcal{G}(37\#)$ and our pair structures (309, 225 and 153 cells).

2. The populations of gaps in $\mathcal{G}(53\#)$ obtained from $n_{g,j}(47\#)$ by one uniform step agree with an independently computed profile of $\mathcal{G}(53\#)$ for all even $g \le 104$ (52 of 52).

3. The leading coefficient of Holt's model, $\ell_1 = \sum_j w_{g,j}(47\#)$, where $w_{g,j}(47\#)$ is $n_{g,j}(47\#)$ divided by the number of gaps 2 in $\mathcal{G}(47\#)$, which is $\prod_{3\le p\le 47}(p-2)$, equals $\mathfrak{S}(g)/\mathfrak{S}(2) = \prod_{q\mid g,\ q>2} \frac{q-1}{q-2}$, the Hardy–Littlewood singular series $\mathfrak{S}(g)$ normalised by its value for $g = 2$, so that its universal factor $2C_2$ cancels [4, eq. (1.3)], [10]; the equality holds exactly, as a rational number, for all eleven gaps. By Holt's Theorem 1.1 [4] (Corollary 1.2 of [3]) this must hold whenever the aggregate counts $\sum_j n_{g,j}(47\#)$ are right, so this check concerns the totals; the distribution over lengths is checked by the first two items.

Table 1 lists the first coefficients $\ell_j = \sum_{i\ge j} \binom{i-1}{j-1} w_{g,i}(p_0\#)$ of Holt's expansion [4, eq. (1.2)], with p_0 as in the table; the complete matrices are in the accompanying data, and all nonzero coefficients ℓ_j for $2 \le g \le 150$ are listed in Appendix A. Evaluating the models shows how slowly large gaps approach their limits: at $p \approx 10^5$ the relative populations of the gaps 84–104 have reached only between 0.4% and 1.2% of ℓ_1, in line with the parameter $\lambda = \Theta(1/\ln p)$ of [4].

Gaps $106 \le g \le 150$. For larger gaps the initial conditions are taken at later stages, in each case with $g < 2p_1$: $p_0 = 53$ for $g \le 116$, $p_0 = 59$ for $g = 118, 120$, $p_0 = 61$ for $122 \le g \le 132$, $p_0 = 67$ for $134 \le g \le 140$, $p_0 = 71$ for $g = 142, 144$ and $p_0 = 73$ for $146 \le g \le 150$. These matrices were computed with the second method of Section 7 and checked in three ways. For $106 \le g \le 112$ the two methods give identical matrices. For every gap the sum over all lengths equals the number $\prod_{p\le p_0}(p - \nu_p(\{0, g\}))$ of pairs at distance g in $\mathcal{G}(p_0\#)$, which is the identity for ℓ_1 below, and the moments $\sum_j \binom{j-1}{k} n_{g,j}(p_0\#)$ for $k \le 4$ agree with Theorem 3.7, evaluated independently. At every stage the gaps 20, 40 and 60 agree with $\mathcal{G}(47\#)$ carried forward by uniform steps.

g	p_0	$\ell_1 = \mathfrak{S}(g)/\mathfrak{S}(2)$	ℓ_2	ℓ_3
84	47	2.400000	25.8583	129.4360
86	47	1.024390	11.4282	59.3230
88	47	1.111111	12.5253	65.7569
90	47	2.666667	31.1566	169.9885
92	47	1.047619	12.6201	71.1408
94	47	1.022222	12.4851	71.4527
96	47	2.000000	24.9092	145.4980
98	47	1.200000	15.3165	91.7893
100	47	1.333333	17.2173	104.4522
102	47	2.133333	28.4414	178.3882
104	47	1.090909	14.8365	95.0951
106	53	1.019608	13.7526	87.4108
108	53	2.000000	27.4986	178.3419
110	53	1.481481	21.0815	141.7996
112	53	1.200000	17.1466	115.7727
114	53	2.117647	31.0361	215.3832
116	53	1.037037	15.5171	110.0515
118	59	1.017544	14.9431	103.9436
120	59	2.666667	40.4254	290.6921
122	61	1.016949	15.5665	113.1496
124	61	1.034483	15.8047	114.6004
126	61	2.400000	37.7316	281.9690
128	61	1.000000	16.0541	122.5834
130	61	1.454545	23.5718	181.8048
132	61	2.222222	36.9175	292.2726
134	67	1.015385	16.7839	132.2527
136	67	1.066667	17.9720	144.4877
138	67	2.095238	35.5301	287.5119
140	67	1.600000	27.8547	231.6899
142	71	1.014493	17.5423	144.9346
144	71	2.000000	35.1447	295.2872
146	73	1.014085	17.9333	151.7206
148	73	1.028571	18.0446	151.4317
150	73	2.666667	48.0656	414.7880

Table 1: Coefficients of Holt's exact models for the gaps $84 \le g \le 150$, with the stage p_0 of the initial conditions.

Holt's worked example. For the constellation $s = 2, 10, 2, 10, 2$ ($|s| = 26$) Holt lists the populations of s and its driving terms of lengths $5, 6, 7$ at $p = 5, 7, 11, 13$ [4, Fig. 6]: $(0, 0, 1)$, $(1, 0, 2)$, $(6, 4, 8)$, $(52, 44, 48)$, and notes that the uniform system holds from $p = 13$ on. The three transitions $5 \to 7$, $7 \to 11$ and $11 \to 13$ lie outside his model ($|s| \ge 2q$). Theorem 3.4 derives each of them exactly from the previous stage; they involve 3, 3 and 18 removals of several points in one image, respectively.

A remark on quadratic density. Holt shows that the expected quadratic density of a gap increases along the sieve [4, Thm. 4.3], under the assumption $g < 2p_1$. The proof uses the recursion only through the inequality $n_{g,1}(p_k\#) \ge (p_k - 2)\, n_{g,1}(p_{k-1}\#)$. By Theorem 4.1 this inequality holds for every gap, since $n_{g,1}(qN) = (q - 2 + e_g)\, n_{g,1}(N) + B_g^{(q)}(N)$ with $e_g \ge 0$ and $B_g^{(q)}(N) \ge 0$. Hence the conclusion of [4, Thm. 4.3] holds for every gap once it occurs, without the restriction $g < 2p_1$. Holt states the result for every gap in [16, Thm. 3.3], with a proof through the same inequality in the uniform regime; Theorem 4.1 supplies this inequality for all spans.

6.2 Ziller's table of missing differences

Ziller [5] tabulated, for $k \le 44$, the even numbers below $h(k)$ that do not occur as gaps in $\mathcal{G}(p_k\#)$ — the *missing differences* (missing gaps, in Holt's terminology) of $\mathcal{D}(k)$. Writing $N_{\min}(k)$ for the largest even number such that all even numbers up to it belong to $\mathcal{D}(k)$, he conjectured that $h(k-1) \le N_{\min}(k)$ for $k > 1$ [5, Conj. 4.1]: every even number up to $h(k-1)$ occurs as a gap of $\mathcal{G}(p_k\#)$, or equivalently, every missing difference of $\mathcal{D}(k)$ exceeds $h(k-1)$. We reproduced his table independently for $18 \le k \le 30$ by an exhaustive search (Section 7); all entries agree.

A missing difference of $\mathcal{D}(k)$ arises from $\mathcal{G}(p_{k-1}\#)$ with $q = p_k$, and Corollary 4.6 classifies it. For $k \le 19$ the calm regime applies ($h(k-1) < 2p_k$), so all missing differences are delays and holes by Corollary 4.4. For $20 \le k \le 30$ we searched $\mathcal{G}(p_{k-1}\#)$ exhaustively for the constellations $(2q, g-2q)$ whose existence would make g a blocked birth; none exists, so all are delays. Ten of these fourteen negative results were confirmed independently with a SAT model solved by CP-SAT; four of them $(170, 230, 256, 296)$ are impossible already modulo 3. The position (hole or shadow) is determined by the reach of pairs $M_2^{(q)}$, obtained from a complete classification of the new differences for $20 \le k \le 27$ (all 54 present differences in these bands: 50 arise from pairs and 4 only coincidently, namely 190, 216, 234 and 282, all of them maxima; at least nine of the 50, namely 200, 202, 206, 210, 224, 226, 268, 274 and 280, also arise coincidently), from the pair sum 326 at $k = 30$, and from the type of the maxima (Section 6.3). Witnesses for every value of $M_2^{(q)}$ in Table 2 and for all 54 births are in the accompanying data.

k	p_k	missing g	$M_2^{(q)}$	$h(k)$, birth	basis
6	13	20	22	22, pair	theorem
8	19	32	34	34, pair	theorem
14	43	86, 88	90	90, pair	theorem
15	47	98	100	100, pair	theorem
19	67	146	152	152, pair	theorem
20	71	166, 170, 172	174	174, pair	search + SAT
21	73	182; **186, 188**	184	190, coincident	search + SAT
22	79	194	200	200, pair and coincident	search + SAT
24	89	230	232	234, coincident	search + SAT
25	97	254, 256	258	258, pair	search; SAT for 256
27	103	278	280	282, coincident	search
28	107	296, 298	300	300, pair	search; SAT for 296
30	113	**328**	326	330, coincident	search

Table 2: All missing differences of $\mathcal{D}(k)$ for $k \le 30$. All are delays. Shadows in bold; all others are holes. "Theorem": calm regime; "search": exhaustive search for the blocking constellation; "SAT": independent confirmation by CP-SAT (for 254 the solver did not finish within one hour).

Result. All 20 missing differences up to $k = 30$ are delays; none is caused by a blocked birth. Seventeen are holes in the set of qualifying pair sums, and three — 186, 188 at $k = 21$ and 328 at $k = 30$ — lie in the shadow of a maximum produced by a coincident birth. Blocked births do occur as individual events, but in this range they never remove a difference from the table.

Observations in other sieves. Replacing the primorial chain by sieves built from random subsets and orders of the primes up to 37, we examined 3689 such sieves (moduli up to $3 \cdot 10^7$, computed by brute force; the survey uses a fixed random seed and is reproducible). The birth law was exact in all of them; among their 734 missing differences, 356 were shadows, none

was caused by a blocked birth, and every one of them was born at the next stage. These are observations on small moduli, not evidence about large k.

6.3 How maximal gaps arise

By Corollary 4.7, a maximal gap of $\mathcal{G}(p_n\#)$ arises from $\mathcal{G}(p_{n-1}\#)$ either by a pair birth, in which p_n alone removes one point of the maximal interval, or by a coincident birth, in which it removes two points at distance $2p_n$ (up to $n = 64$ one has $h(n) < 4p_n$, so no other route is possible). Counting the points removed by p_n alone in maximal intervals classifies the route. We used the complete lists of maximal configurations for $n \le 54$ published by Ziller and Morack [9] (ancillary file `remainders.txt`) and the start positions of one maximal interval for each $n \le 64$ from the OEIS table of A048670 (R. Gerbicz; the values for $n = 62$–64 computed by A. Bożek) [12]; see also [6, 7].

- For $n \le 19$ all maxima are pair births, as the calm regime requires; $n = 20$ is a pair birth as well.
- For $21 \le n \le 54$ (all configurations): coincident births only for 22 values of n, pair births only for 6 ($n = 25, 28, 29, 31, 35, 42$), both routes for 6 ($n = 22, 26, 33, 34, 38, 46$).
- For $55 \le n \le 64$ (one interval each): coincident births at $n = 56, 58, 59, 61, 62, 64$ and pair births at $n = 55, 57, 60, 63$.
- In every coincident birth the two points are exactly $2p_n$ apart; the two sources agree wherever both are available.

Thus from $n = 21$ on both routes occur, the coincident one more often. The data do not support a simple rule deciding the route: an extrapolation from $n \le 54$, where $47 \le n \le 54$ are all coincident births, predicted coincident births for $n = 55$–57 and failed at $n = 55$ and 57; nor does the ratio $2p_n/h(n-1)$ decide it, since pair births occur even at its smallest values in the range.

6.4 The survival hypothesis: a test up to 10^9

Below p_{k+1}^2 the survivors of the sieve by the primes up to p_k are the primes, so the gaps between primes in the interval of survival $[p_k^2, p_{k+1}^2)$ are gaps of $\mathcal{G}(p_k\#)$. Holt's survival hypothesis [16, Conj. 2.1] states that these gaps form an approximately uniform sample of $\mathcal{G}(p_k\#)$. That the frequencies of small gaps between consecutive primes agree with predictions from the Hardy–Littlewood conjecture is classical [13, 17]; the question here is at which stage of the cycle the sample is taken, with exact populations of the cycle in place of asymptotic constants. With the models of Section 6.1 this can be tested for all gaps $g \le 104$ and all primes from 47^2 to 10^9: 50 738 584 gaps of size at most 104 in 3387 intervals. For each interval we multiply its number of gaps by the share $n_{g,1}/\varphi$ of g in the cycle at a stage z, obtained from 47# by Holt's chain (exact for $g \le 104$ and $q \ge 53$; the models of Section 6.1 for $106 \le g \le 150$ start at later stages and are not used here), and compare the sums E_g with the actual counts A_g (Table 3; $x = p_k^2$, totals normalised to the same number of gaps $g \le 104$).

At $z = \sqrt{x}$ the cycle has too many small gaps and too few large ones: $A_2/E_2 - 1 = -10.9\%$, $A_{104}/E_{104} - 1 = +121\%$. By Mertens' theorem the mean gap of the cycle at $\sqrt{x}$ is about $e^\gamma \log\sqrt{x} \approx 0.89 \log x$, and for $g = 2$ the Hardy–Littlewood conjecture predicts the ratio $e^\gamma/2 = 0.8905$; this is the paradox discussed by Pólya [18]. At $z = x^{1/e^\gamma}$, the level of the random model of Banks, Ford and Tao [14], at which the density of the cycle matches $1/\log x$, the deviations are at the level of sampling noise, and among the exponents tested this one minimises them. The same holds for constellations. For all constellations of length 2 and 3 and span at most 40 — 141 patterns of pairs and 552 of triples of consecutive gaps, 30 891 159 pairs

stage $z = x^c$	median of $\lvert A_g - E_g\rvert / E_g$	χ^2 per degree of freedom
$c = 0.50$ $(\sqrt{x})$	34.17%	19316.8
$c = 0.53$	14.76%	4313.9
$c = 0.55$	4.81%	511.7
$c = 0.56$	0.46%	6.8
$c = e^{-\gamma} = 0.5615$	0.11%	0.7
$c = 0.565$	1.48%	52.7
$c = 0.57$	3.40%	281.2
$c = 0.60$	13.26%	4916.9

Table 3: Gaps $2 \le g \le 104$ between primes in $[47^2, 10^9]$ against the cycle at stage x^c.

and 14 880 619 triples between 47^2 and 10^9 — we take the initial conditions from a complete enumeration of $\mathcal{G}(19\#)$ and apply Holt's chain from $q = 23$ on, which is exact since $2 \cdot 23 > 40$ (Table 4). In $\mathcal{G}(p\#)$ a constellation and its reversal have equal populations; among the primes up to 10^9 the counts of (a, b) and (b, a) differ by less than two standard deviations for the ten pairs we compared, for example 379 483 and 379 720 for $(2, 4)$ and $(4, 2)$. The comparison is empirical; nothing is proved about primes.

	pairs		triples	
stage $z = x^c$	median dev.	χ^2/d.f.	median dev.	χ^2/d.f.
$c = 0.50$ $(\sqrt{x})$	7.18%	1082.5	4.59%	79.1
$c = 0.55$	1.10%	30.9	0.80%	3.1
$c = 0.56$	0.20%	1.4	0.45%	1.0
$c = e^{-\gamma}$	0.14%	1.0	0.44%	1.0
$c = 0.565$	0.31%	3.7	0.49%	1.2
$c = 0.57$	0.77%	16.4	0.66%	2.2

Table 4: Pairs and triples of consecutive gaps of span at most 40 between primes in $[47^2, 10^9]$, against the cycle at stage x^c.

Statistics. For each category — a gap, or a pattern of two or three consecutive gaps — A is its total count over all intervals and E the sum over the intervals of the number of gaps in the interval times the share of the category in the cycle at stage $z = x^c$. The values E are rescaled so that the totals of A and E agree, which leaves 51, 140 and 551 degrees of freedom for the gaps, the pairs and the triples. Consecutive gaps are not independent, so χ^2 per degree of freedom is used as a measure of fit on the scale of Poisson noise, not as a formal test. To check this scale we computed the same statistic for the count of each gap in each interval separately, over the 140 730 cells with $E \ge 5$ in 3387 intervals: it is 0.90 at $c = e^{-\gamma}$ and 8.37 at $c = 1/2$. In our data, at the level x^{1/e^γ} the counts thus scatter about the model at the scale of independent sampling, and the dependence between consecutive gaps does not inflate the scatter.

Holt also asks for the largest gaps inside the intervals of survival [16, §4.1]. Up to 10^9 the largest gap lying in $[p_k^2, p_{k+1}^2)$ is at most 0.515 times the threshold $2p_k + 4$ relevant for Legendre's conjecture (at $p_k = 31$), and exceeds half of it only there, among the 3401 intervals up to 10^9 counted from $p_k = 2$; counting instead the gaps that start in the interval gives 0.778 at $p_k = 7$, from the gap from 113 to 127; for $k \le 64$ it is an order of magnitude below Jacobsthal's function, for example 42 against 1110 at $p_k = 311$.

7 Computations and verification

Programs. The population matrices were computed by an exact counting engine that enumerates the windows of a wheel of small primes and treats the remaining primes by an exact covering computation; its results agree with Holt's published data for $\mathcal{G}(37\#)$ (415 cells) and with his recursion across $37\# \to 41\# \to 43\#$ (408 cells). Ziller's table was verified for $18 \le k \le 30$ by an exhaustive search over the sectors of a wheel, with every positive answer certified by an explicit witness checked with gcd. The same machinery, restricted to prescribed interior points, decides the existence of constellations for the classification of Section 6.2. The independent check models the existence of a constellation as a covering problem — one residue class for each prime, all interior points except the prescribed ones covered, the prescribed points and the endpoints not covered — solved by Google OR-Tools CP-SAT; the model was validated against brute force on 70 constellations in $\mathcal{G}(19\#)$.

Brute-force verification. Independently of the engine, all statements of Sections 3–5 were checked by listing complete cycles for moduli up to 510510, including sieves that omit a small prime, in which multi-point removals are frequent. Table 5 summarises the checks.

Statement	Test	Result
Thm. 3.4, gaps	full cycles; $7\# \to 11$, $11\# \to 13$, $13\# \to 17$ and sieves without 5, 7, 11	1585 cells, 0 errors; classes up to 5 points
Thm. 3.4, gaps, engine	$23\# \to 29\#$, $37\# \to 41\#$, $41\# \to 43\#$, $43\# \to 47\#$	805 cells, 0 errors
Thm. 3.4, constellations	160 constellations, spans $2q$–$5q$	425 cells, 0 errors
Cor. 3.6	same tests	first 2-, 3-, 4-, 5-point classes at $2q, 6q, 8q, 12q$
Thm. 3.7	$k = 0, \dots, 3$, 180 constellations; engine for gaps	720 of 720; 609 cells
Thm. 4.1	all births of constellations with at most 4 gaps	783 of 783
Lemma 4.2, Cor. 4.3	chain $2 \to \cdots \to 30030$, 389 constellations	no exception; 1945 of 1945
Thm. 5.1	180 + 600 constellations	sufficient condition never violated
Convention $\lvert s\rvert < N$	spans $\ge N$, including $N = 2$	laws hold (214 + 60 cases)

Table 5: Computational verification.

Exhaustive verification (version 3). The shape identity

$$m_P(qN) = \big(q - \nu_q(P)\big)\, m_P(N) + \sum_K m_{P \cup K}(N),$$

where K runs over the sets that form a full residue class modulo q of $P \cup K$ and contain no endpoint, is Theorem 3.4 for all constellations at once. It was checked for every shape of span at most 40 on complete cycles, and Theorems 4.1, 5.1 and 5.2 for every constellation of span at most 40 (Table 6). In all 243 117 cases with a populated shape having two congruent points, the excess at length J^* equals the sum in Theorem 5.2.

Two methods for $n_{g,j}$. The matrices are computed window by window: a window is a pair of survivors of a small base wheel at distance g, and each prime above the base removes one residue class. The first method counts the assignments of classes by dynamic programming

N	q	shapes	constellations	chain exact / inexact
$2\cdot3\cdot5\cdot7$	11	1247	7823	469 / 7354
$2\cdots11$	13	3889	7823	979 / 6844
$2\cdots13$	17	6696	7823	3224 / 4599
$2\cdots17$	19	7734	7823	6376 / 1447
$2\cdot5\cdot7\cdot11\cdot13$	3	12431	171568	3 / 171565
$2\cdot3\cdot7\cdot11\cdot13$	5	5615	13398	9 / 13389
$2\cdot3\cdot5\cdot11\cdot13$	7	4242	8678	90 / 8588
$2\cdot3\cdot5\cdot7\cdot13$	11	3889	7823	469 / 7354
$2\cdot3\cdot7\cdot11\cdot13\cdot17$	5	10601	13398	9 / 13389
$2\cdot3\cdot5\cdot11\cdot13\cdot17$	7	7387	8678	90 / 8588
total		63731	254835	

Table 6: Exhaustive verification, span at most 40: all shape identities and all birth laws hold; Holt's chain is exact exactly when no populated shape has two congruent points.

over the sets of nodes still uncovered. The second applies Theorem 3.7 to a single window: the number of assignments that leave a given set T of nodes uncovered is $\prod_p(p-\nu_p(\{0,g\}\cup T))$, the product over the primes above the base, as in Theorem 3.7, and binomial inversion of the sums over $|T|=k$ gives the number of assignments leaving exactly m nodes uncovered. It uses the symmetry $x\mapsto -x$ and exact integer arithmetic, with sums modulo several primes and the Chinese remainder theorem where products exceed 2^{63}. The two methods agree on the whole matrices for $106\le g\le 112$ in $\mathcal{G}(53\#)$ and on single windows up to $79\#$; the second is several hundred times faster per window.

Data availability. Code, data (the matrices $n_{g,j}$ for $41\#$, $43\#$, $47\#$, the pair structures, the classification results and the verification logs) and the programs used for all computations are available at `https://github.com/hojka1987t-dot/sieve-dynamics-beyond-2p`; the version used in this paper (release v2.0.0) is archived at Zenodo under doi:10.5281/zenodo.23009304 (all versions). Table 1, Appendix A and the census of Section 6.3 are reproduced by the scripts in `code/analysis`, Table 5 and the survey of Section 6.2 by those in `code/brute_force`. The classification in Table 2 rests on the result files in `data/`, produced by the exhaustive searches and the SAT check, whose programs are included; these runs take from hours to about a day. The exhaustive verification of Section 7, the survival test of Section 6.4, the computations of Part II are reproduced by the scripts in `code/exhaustive`, `code/survival` and `code/windows` of release v2.0.0; the matrices for $106\le g\le 150$ and the second method are in `data` and `code/engine_45`.

8 Minimal windows and the depth of their ancestors

For $y\ge 0$ let $W_N(y)$ be the least number of points of $\Omega(N)$ in $y+1$ consecutive integers; this is $\varphi_{\min}(y+1,\cdot)$ of Costello and Watts [15] and the minimum of the interval sieve in [14]. For a window w of length y let $|w|$ be its number of points of $\Omega(N)$ and $c_q(w)$ the largest number of them in one residue class modulo q.

Lemma 8.1. *$W_{qN}(y)=\min_w\big(|w|-c_q(w)\big)$, over all windows w of length y.*

Proof. By Lemma 3.2, the q translates of w by multiples of N remove the residue classes of its points modulo q, each class in exactly one translate. □

Write $W = W_N(y)$ and, for $j \geq 0$, $C_j = \max\{\, c_q(w) : |w| = W + j \,\}$. By Lemma 8.1, $W_{qN}(y) = \min_{j \geq 0}(W + j - C_j)$. The least j at which the minimum is attained is the *depth* $d_q(y)$: the minimum for qN arises from windows with $W + d_q(y)$ points.

Theorem 8.2. *Let $2, 3 \mid N$ and $W_N(y) \geq 1$, and let c bound the number of points of a window of length y in one class modulo q.*

(a) If $y < 6q$, then $d_q(y) = 0$, that is, $W_{qN}(y) = W_N(y) - C_0$.

(b) If $y < 8q$, then $d_q(y) \leq 1$.

(c) In general $d_q(y) \leq \max(0,\, c - C_0 - 1)$.

Proof. A window with $W + j$ points gives a value below $W - C_0$ only if $C_j \geq C_0 + j + 1$, hence only if $j \leq c - C_0 - 1$; this is (c). Since $W \geq 1$, $C_0 \geq 1$. For $y < 6q$ a class has at most two points (Corollary 3.6), and (c) with $c = 2$ gives (a); for $y < 8q$ at most three, by the argument following Corollary 3.6, and (c) with $c = 3$ gives (b). □

The case $W_{qN}(y) = 0$ is Corollary 4.7: the maximal gaps of $\mathcal{G}(qN)$ arise from windows whose points form one class modulo q.

9 Computations on minimal windows

A certified recursion. Theorem 8.2(c) turns Lemma 8.1 into an algorithm. Given $W_N(y)$, compute $C_0, C_1, \ldots$ by exhaustive search among windows with $W+j$ points, and stop as soon as $W+j+1-c$ is at least the current minimum, where c is an upper bound for the size of a class: the largest number of survivors among $\lfloor y/(2q) \rfloor + 1$ consecutive terms of an arithmetic progression with difference $2q$, computed exactly for at most 17 terms and bounded by subadditivity beyond. The output is $W_{qN}(y)$ with a certificate, independent of any assumption on the depth. Table 7 compares it with a direct branch-and-bound search over residue classes; the whole profile for $q = 79$, $198 \leq y \leq 700$ (252 lengths), took 5.9 hours of processor time.

y	$W_{qN}(y)$	direct search, all cores	certified recursion, one core
300	9	12.2 s	0.4 s
400	19	12.8 s	1.5 s
500	29	157.4 s	47.0 s
600	39	118.2 s	5.7 s
700	50	192.9 s	56.6 s

Table 7: $N = 73\#$, $q = 79$; both methods give the same values.

Depth at most one up to about $20q$. For $N = p_k\#$, $7 \leq k \leq 21$, $q = p_{k+1}$ and all even y with $W_N(y) \geq 1$ in the ranges of Table 8, 3943 values in total, $d_q(y) \leq 1$. For $N \leq 23\#$ all windows were enumerated; for $N = 73\#$ the values were computed by the certified recursion, and the depth is 0 at all 252 lengths. For $29\# \leq N \leq 71\#$, W_N and W_{qN} were computed by branch and bound; in 25 cases in which the stored optimum came from a deeper window, an exhaustive search exhibited an optimal window of depth at most one. Depth 1 is needed, for instance, for $N = 29\#$ at $y = 630, 632$; $N = 41\#$ at $y = 556, 566, 590, 598$; $N = 43\#$ at $y = 394, 402, 442, 448, 476, 618, 628, 634, 642$; and $N = 59\#$ at $y = 682$. In all these cases $C_1 = C_0 + 2$.

N	q	y	values	N	q	y	values
17#	19	26–200	88	43#	47	98–700	302
19#	23	38–400	182	47#	53	104–700	299
23#	29	44–400	179	53#	59	116–700	293
29#	31	56–700	323	59#	61	130–700	286
31#	37	64–700	319	61#	67	150–700	276
37#	41	72–700	315	67#	71	172–700	265
41#	43	88–700	307	71#	73	188–700	257
				73#	79	198–700	252

Table 8: Ranges in which $d_q(y) \le 1$ was verified.

Long windows. Table 9 gives the depth for long windows, by enumeration of all windows for $N = 11\#, 13\#, 17\#$ and by the certified recursion along $13\# \to 17\# \to 19\# \to 23\# \to 29\# \to 31\# \to 37\#$. Depth 2 occurs sporadically and in clusters, first at about $39q$ to $217q$ depending on N; depth 3 did not occur, also in windows up to $y = 3000$, that is up to about $230q$ for $N = 11\#$. In all 35 cases of depth 2 for $N = 11\#, 13\#, 17\#$ and $y \le 3000$ we have $C_2 = C_1 + 2$. Depth at least 1 is not associated with increases of the capacity c (difference in frequency -0.7 percentage points, $p = 0.63$ in a test by cyclic shifts, which preserves the dependence between nearby lengths), but it is associated with the scarcity of optimal windows: for $N = 11\#, 13\#, 17\#$ its frequency falls from 7.5%, 12.9%, 18.5% in the lowest quartile of the number of optimal windows to 1.9%, 5.0%, 8.4% in the highest ($p < 0.001$ in the same test). A plausible reading is that when the minimum is attained by few windows, the largest class among them is a maximum over a small set, while the windows with one or two more points are far more numerous. For $N = 17\#$, $q = 19$, $y = 820$: $W_N(y) = 141$, $W_{qN}(y) = 131$ and $(C_0, C_1, C_2) = (9, 10, 12)$, so the minimum 131 is attained only by windows with $W + 2$ points.

method	$N \to qN$	y	$d = 0$	$d = 1$	$d = 2$ (first y)
enumeration	$11\# \to 13$	26–3000	1416	65	7 (2820)
enumeration	$13\# \to 17$	34–3000	1342	131	11 (1144)
enumeration	$17\# \to 19$	38–3000	1279	186	17 (814)
recursion	$13\# \to 17$	700–1500	170	29	2 (1144)
recursion	$17\# \to 19$	700–1500	186	13	2 (820)
recursion	$19\# \to 23$	700–1500	174	27	—
recursion	$23\# \to 29$	700–1500	196	5	—
recursion	$29\# \to 31$	700–1500	188	13	—
recursion	$31\# \to 37$	700–1500	194	6	1 (1428)

Table 9: Depth for long windows (enumeration: all even y; recursion: $y \equiv 0 \bmod 4$).

10 Open questions

1. *Ziller's Conjecture 4.1.* By Theorem 4.1 it states that every even number up to $h(k-1)$ is a gap of $\mathcal{G}(p_{k-1}\#)$ or the span of a qualifying constellation; in the calm regime, a gap or a sum of two adjacent gaps. Is there a structural reason why the holes of this set stay near its top? In all small sieves we examined no missing difference persisted for more than one stage. By Corollary 4.6 a missing difference stays absent at the next stage exactly when no qualifying constellation produces it, and up to $k = 30$ every missing difference is a delay (Section 6.2).

2. *Which route produces the maximum?* Both pair and coincident births of maxima occur for $n \geq 21$, with no simple rule deciding between them.

3. *Depth.* Is the depth $d_q(y)$ of Section 8 bounded? It is 0 below $6q$ and at most 1 below $8q$; in all computed cases, with windows up to about $230q$, it never exceeds 2, while Theorem 8.2(c) gives only a bound that grows with the capacity c. Equivalently, $d_q(y) \leq 2$ exactly when $\max_j(C_j - j)$ is attained at some $j \leq 2$.

4. *Higher thresholds.* Up to $n = 64$ the maxima stay below $4p_n$. For larger n they will pass the thresholds $4p_n$ and $6p_n$, where births through pairs at distance $4q$ and through classes of three points become possible for maximal gaps.

5. *Survival at the level x^{1/e^γ}.* The agreement of Section 6.4 holds for gaps up to 104 and for pairs and triples of span at most 40. Does it persist for longer constellations, larger spans and larger x, and what is the size of the remaining deviation?

6. *Paired progressions.* The same method applies to sieves removing two residue classes per prime, as in the paired Jacobsthal function of Ziller and Morack [11]; the classes of coincidence and the thresholds then change.

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Note on versions. Version 3 adds Theorem 5.2, the exhaustive verification in Section 7, the extension of Holt's exact models to $g \leq 150$ with a second exact method, the complete coefficients in Appendix A, the survival test in Section 6.4 and Part II. It states the classification of the new differences in Section 6.2 more precisely, with archived witnesses, proves the thresholds $8q$ and $12q$ stated without proof in version 2, restricts the statement on shadows in Corollary 4.6 to what is proved, attributes the threshold $6q$ to Haugland and Hagedorn and related work to Samuel and Tucker, and adopts the terms uniform and non-uniform regime suggested by Holt. No numerical result of version 2 has changed. Version 2 attributed the residue-class criterion to Holt [3, Lemma 3.1], and the identity for ℓ_1 to [4, Thm. 1.1] and [3, Cor. 1.2].

A All coefficients of Holt's models

For each even $g \leq 150$ the coefficients $\ell_j = \sum_{i \geq j} \binom{i-1}{j-1} w_{g,i}(p_0\#)$, $j = 1, 2, \ldots$, of Holt's expansion [4, eq. (1.2)], with $w_{g,i}(p_0\#) = n_{g,i}(p_0\#)/\prod_{3 \leq p \leq p_0}(p-2)$, where $p_0 = 47$ for $g \leq 104$ and as in Section 6.1 for larger g; eight significant digits. Exact values are in the file `wspolczynniki_1.csv` of the accompanying data.

g	$\ell_1, \ell_2, \ldots$
$p_0 = 47$	
2	1
4	1
6	2, 0.60536573
8	1, 0.60536573, 0.061719984
10	1.3333333, 0.90804859, 0.12343997
12	2, 2.1187801, 0.61719984, 0.042479193
14	1.2, 1.5134143, 0.57605318, 0.06371879

g	$\ell_1, \ell_2, \ldots$
16	1, 1.5134143, 0.74063981, 0.12743758, 0.0051352247
18	2, 3.4808529, 2.0573328, 0.46727112, 0.030811348
20	1.3333333, 2.9511579, 2.386506, 0.84958386, 0.12324539, 0.0045578743
22	1.1111111, 2.3836276, 1.8574776, 0.61948823, 0.073947236
24	2, 5.1456087, 5.1080634, 2.4354737, 0.55460427, 0.04785768
26	1.0909091, 3.1630359, 3.6215588, 2.0748431, 0.61974826, 0.090397839, 0.005019041
28	1.2, 3.4997706, 4.0206161, 2.3009563, 0.67990375, 0.095715359, 0.0047800391
30	2.6666667, 9.1334555, 12.722154, 9.2409945, 3.7231846, 0.81377047, 0.086399206, 0.0032742137
32	1, 3.8440724, 6.1501188, 5.3081292, 2.6783865, 0.80028676, 0.13664937, 0.01207821, 0.00042594272
34	1.0666667, 4.1348641, 6.6752536, 5.8099146, 2.9470054, 0.8780175, 0.14685342, 0.012278301, 0.00038943335
36	2, 8.401071, 14.876199, 14.441783, 8.3726838, 2.9622574, 0.62816353, 0.075507006, 0.0046123512, 0.00011090015
38	1.0588235, 4.727528, 8.9314067, 9.2781005, 5.7580305, 2.1661266, 0.47594704, 0.054654812, 0.0024136754
40	1.3333333, 6.1563533, 12.119273, 13.251886, 8.7820879, 3.6071397, 0.89941747, 0.12696673, 0.008664892, 0.00019011454
42	2.4, 12.1992, 26.806448, 33.310172, 25.691824, 12.694319, 3.9963698, 0.76928833, 0.082440876, 0.0040127041, 5.2400275e-05
44	1.1111111, 5.9846078, 14.068599, 18.943976, 16.114924, 9.0030567, 3.3237398, 0.79373943, 0.11592606, 0.0091819382, 0.000290875
46	1.047619, 5.8455628, 14.261662, 19.958036, 17.65539, 10.251105, 3.9223411, 0.96452391, 0.14308326, 0.011175335, 0.00032431785
48	2, 11.635454, 29.766332, 44.007736, 41.551494, 26.126049, 11.061326, 3.1144819, 0.5614013, 0.060022313, 0.0032685847, 6.2591641e-05
50	1.3333333, 8.2821599, 22.846766, 36.862979, 38.555368, 27.366443, 13.40668, 4.5170226, 1.0225499, 0.14809011, 0.012507593, 0.00050914247, 5.7985346e-06
52	1.0909091, 6.9479771, 19.609536, 32.27344, 34.28374, 24.563124, 12.035664, 3.9990672, 0.87250513, 0.11692329, 0.0084119779, 0.00023334647
54	2, 13.246699, 39.232218, 68.531432, 78.406568, 61.67592, 34.053822, 13.224923, 3.5598985, 0.64167623, 0.072571206, 0.0045392032, 0.00011558412
56	1.2, 8.3521553, 26.09316, 48.311875, 58.956029, 49.890671, 29.996273, 12.915065, 3.9631392, 0.85214243, 0.12432913, 0.011631194, 0.00062438621, 1.4407892e-05
58	1.037037, 7.2115024, 22.478413, 41.439065, 50.195138, 41.966748, 24.752242, 10.340414, 3.0266436, 0.60444168, 0.07861726, 0.006172424, 0.00025713537, 4.2357696e-06
60	2.6666667, 19.931526, 67.194473, 134.97372, 179.74255, 167.03385, 111.0166, 53.191289, 18.276545, 4.4226467, 0.7290003, 0.077548386, 0.0048880217, 0.00015831633, 1.8831171e-06
62	1.0344828, 8.2527024, 29.880419, 64.936954, 94.386781, 96.769568, 71.903499, 39.161708, 15.625232, 4.5141294, 0.9214869, 0.1274249, 0.011109595, 0.00053441888, 1.0299754e-05
64	1, 7.8406088, 27.845779, 59.215039, 83.966421, 83.659293, 60.106774, 31.441965, 11.937585, 3.2385898, 0.60878626, 0.075233261, 0.005593282, 0.00021410403, 3.1474957e-06
66	2.2222222, 18.428738, 69.584964, 158.28998, 241.86447, 262.00259, 206.95826, 120.73997, 52.105803, 16.495161, 3.7590821, 0.59745972, 0.063043818, 0.0040947145, 0.00014504205, 2.1810603e-06
68	1.0666667, 9.245533, 36.624107, 87.780389, 142.03676, 163.9083, 138.90404, 87.68133, 41.365736, 14.49746, 3.7149344, 0.67666231, 0.083665625, 0.0065050376, 0.0002763833, 4.5879864e-06
70	1.6, 14.048144, 56.427667, 137.31004, 225.92177, 265.61912, 229.91255, 148.71942, 72.21372, 26.207155, 7.014813, 1.3526331, 0.18109058, 0.015933832, 0.00085308712, 2.5407837e-05, 3.8016317e-07

g	$\ell_1, \ell_2, \ldots$
72	2, 18.515244, 78.705236, 203.53084, 357.58697, 451.42755, 422.30648, 297.54446, 158.87162, 64.155523, 19.400434, 4.3116349, 0.68344768, 0.073791334, 0.0050501421, 0.00019383397, 3.1683496e-06
74	1.0285714, 9.6324371, 41.488999, 108.92364, 194.72215, 250.78545, 240.07184, 173.6911, 95.607302, 39.972457, 12.569443, 2.9156175, 0.48293986, 0.054160709, 0.0037516384, 0.00013384967, 1.4941787e-06
76	1.0588235, 10.284944, 46.083274, 126.27786, 236.53472, 320.65142, 324.85908, 250.41959, 148.11719, 67.287455, 23.34404, 6.1075762, 1.1807316, 0.1636909, 0.015607805, 0.00096718902, 3.5890668e-05, 6.5002772e-07
78	2.1818182, 21.567728, 98.445945, 275.1367, 526.36265, 729.96793, 758.04676, 600.3778, 365.90001, 171.87326, 61.915465, 16.903677, 3.4278201, 0.50038836, 0.050091559, 0.0031769049, 0.00011028896, 1.4647798e-06
80	1.3333333, 13.722972, 65.37844, 191.23879, 384.10937, 561.24768, 616.59233, 519.09041, 338.18024, 170.97814, 66.867814, 20.044428, 4.5340055, 0.75620794, 0.090075354, 0.0073507782, 0.0003908142, 1.2673587e-05, 2.0550101e-07
82	1.025641, 10.73904, 52.093809, 155.29886, 318.22235, 474.8943, 533.50014, 459.88462, 307.21991, 159.51867, 64.175594, 19.82184, 4.626614, 0.79689221, 0.097934234, 0.0082056875, 0.00044266715, 1.4357216e-05, 2.3721413e-07
84	2.4, 25.858285, 129.43604, 399.46107, 850.55428, 1324.766, 1561.3457, 1420.7726, 1009.458, 562.60857, 245.76064, 83.639859, 21.929755, 4.3553913, 0.64034185, 0.067745337, 0.0050075643, 0.00025300021, 8.4706776e-06, 1.5001684e-07
86	1.0243902, 11.428245, 59.32301, 190.17933, 421.43184, 684.56421, 843.43699, 804.49227, 600.97136, 353.38373, 163.50461, 59.196142, 16.586255, 3.53445, 0.55835651, 0.063045554, 0.0048294211, 0.0002324582, 6.1897956e-06, 6.5897389e-08
88	1.1111111, 12.525314, 65.756852, 213.41963, 479.35672, 790.29415, 989.79354, 961.43057, 732.95951, 440.96449, 209.38197, 78.082953, 22.637023, 5.0186775, 0.83033287, 0.098935389, 0.0080554925, 0.00041327664, 1.1611321e-05, 1.2917351e-07
90	2.6666667, 31.1566, 169.98845, 575.08825, 1350.984, 2338.5136, 3088.638, 3179.9769, 2584.9974, 1670.0587, 858.85629, 350.54248, 112.70014, 28.184565, 5.3818554, 0.76447968, 0.077963186, 0.0054457273, 0.00024506509, 6.474123e-06, 7.5557932e-08
92	1.047619, 12.620139, 71.140821, 249.25194, 608.01491, 1096.1879, 1513.2798, 1635.1842, 1401.864, 960.77656, 527.92171, 232.29518, 81.442764, 22.547354, 4.8635849, 0.80247766, 0.098832761, 0.0087961168, 0.00054008675, 2.0984713e-05, 4.0029878e-07
94	1.0222222, 12.485051, 71.452715, 254.5604, 632.56383, 1164.1958, 1644.6651, 1823.8651, 1610.1714, 1140.9541, 651.2602, 299.3684, 110.37966, 32.389664, 7.4731931, 1.3327704, 0.17950675, 0.017709446, 0.0012301093, 5.7049231e-05, 1.6086381e-06, 2.0569977e-08
96	2, 24.909168, 145.49799, 529.5825, 1346.008, 2537.1219, 3676.4772, 4189.6469, 3809.2817, 2787.4408, 1648.7376, 788.8641, 304.56284, 94.352121, 23.251443, 4.5039359, 0.67532243, 0.076881883, 0.0064782118, 0.00038776226, 1.5058697e-05, 2.892845e-07
98	1.2, 15.316524, 91.789286, 343.17466, 897.04986, 1741.2664, 2601.968, 3061.9781, 2878.9621, 2181.5456, 1337.889, 664.36302, 266.31989, 85.614125, 21.838916, 4.3528225, 0.6633641, 0.074974027, 0.0060111554, 0.00031891187, 9.8593693e-06, 1.2896227e-07
100	1.3333333, 17.217337, 104.45216, 395.62637, 1048.6359, 2066.3003, 3138.6815, 3761.0781, 3608.7704, 2798.4496, 1762.7318, 903.38889, 376.16786, 126.72692, 34.297171, 7.3819017, 1.246796, 0.16245212, 0.01597947, 0.0011531515, 5.8429942e-05, 1.9007006e-06, 3.0533559e-08
102	2.1333333, 28.441362, 178.38816, 699.54694, 1922.52, 3933.5848, 6213.1771, 7752.4395, 7754.7048, 6274.8925, 4126.3694, 2207.2636, 958.04309, 335.41147, 93.798113, 20.655361, 3.5109007, 0.44810305, 0.041332857, 0.0026095792, 0.00010396191, 2.2780153e-06, 2.0164871e-08
104	1.0909091, 14.836529, 95.09509, 381.81951, 1076.6937, 2265.8171, 3690.7808, 4763.1943, 4944.4677, 4167.3964, 2866.3639, 1611.132, 738.59189, 274.65272, 82.073618, 19.431818, 3.5710154, 0.49446347, 0.049364824, 0.0033171391, 0.00013328423, 2.5031663e-06, 8.8983515e-09

$p_0 = 53$

g	$\ell_1, \ell_2, \ldots$
106	1.0196078, 13.752615, 87.410799, 347.98923, 972.85421, 2029.4363, 3276.4949, 4190.5689, 4310.3763, 3599.2386, 2452.1353, 1364.9341, 619.4844, 227.98435, 67.396756, 15.778139, 2.8657341, 0.3920813, 0.038705361, 0.0025825122, 0.00010486262, 2.1646433e-06, 1.6702392e-08
108	2, 27.498621, 178.34191, 725.28473, 2073.9986, 4432.0778, 7343.0893, 9658.0256, 10241.576, 8843.4388, 6253.693, 3629.9335, 1727.9833, 671.96839, 211.91341, 53.589087, 10.69072, 1.6437013, 0.18832239, 0.015288442, 0.0008113541, 2.4336557e-05, 2.9338151e-07
110	1.4814815, 21.081535, 141.79957, 599.43805, 1786.2067, 3988.3462, 6924.9749, 9576.5633, 10716.653, 9805.4391, 7381.6681, 4585.6536, 2350.7854, 991.66712, 342.24866, 95.750935, 21.426401, 3.7629734, 0.50496017, 0.049813512, 0.0034088464, 0.00014747578, 3.4222987e-06, 2.9922776e-08
112	1.2, 17.146559, 115.77271, 491.09921, 1467.7519, 3285.271, 5714.348, 7910.1123, 8851.9762, 8090.0864, 6074.8434, 3757.772, 1914.1332, 800.22482, 272.79385, 75.065433, 16.430422, 2.8017557, 0.3613778, 0.033776404, 0.0021438623, 8.3266309e-05, 1.6536315e-06, 1.2109061e-08
114	2.1176471, 31.036083, 215.38316, 941.23325, 2905.5897, 6737.2139, 12179.858, 17589.588, 20623.767, 19845.485, 15778.813, 10402.696, 5691.1649, 2578.7812, 963.33132, 294.40909, 72.779967, 14.320567, 2.1924155, 0.2527669, 0.020900583, 0.0011464172, 3.6287409e-05, 4.9111997e-07, 1.2492177e-09
116	1.037037, 15.517119, 110.0515, 492.05196, 1556.0768, 3701.619, 6876.9329, 10225.675, 12372.931, 12319.53, 10167.426, 6984.4889, 3999.7545, 1907.7573, 755.39206, 246.845, 65.981645, 14.246567, 2.4418447, 0.32427862, 0.032234324, 0.002278183, 0.00010540562, 2.7451039e-06, 2.8093952e-08
$p_0 = 59$	
118	1.0175439, 14.943082, 103.9436, 455.47887, 1410.5974, 3283.2901, 5962.8461, 8658.5392, 10219.218, 9912.017, 7956.924, 5307.2146, 2944.7521, 1357.4239, 517.82098, 162.38277, 41.444878, 8.4891451, 1.3684844, 0.16895527, 0.015367244, 0.00097204497, 3.9037517e-05, 8.4876865e-07, 7.1066243e-09
120	2.6666667, 40.425405, 290.6921, 1318.8588, 4236.0472, 10244.685, 19371.534, 29354.119, 36247.329, 36891.93, 31181.645, 21984.99, 12955.422, 6378.3173, 2616.5792, 889.88889, 248.96008, 56.663104, 10.332035, 1.4777474, 0.16095045, 0.012791766, 0.00069552537, 2.332119e-05, 4.035197e-07, 2.7172238e-09
$p_0 = 61$	
122	1.0169492, 15.56652, 113.14956, 519.54603, 1691.1053, 4150.8123, 7978.7649, 12313.298, 15517.085, 16154.979, 14004.494, 10158.072, 6179.9785, 3154.2028, 1348.0313, 480.46013, 141.89356, 34.402876, 6.7608513, 1.0582568, 0.12878713, 0.011773639, 0.00076788221, 3.2834099e-05, 7.8373698e-07, 6.9084385e-09
124	1.0344828, 15.804662, 114.6004, 524.61348, 1701.2923, 4157.2681, 7948.9142, 12190.552, 15249.491, 15739.653, 13507.046, 9681.9693, 5809.3122, 2917.2774, 1223.2092, 426.25896, 122.5628, 28.779559, 5.441035, 0.81224853, 0.093182162, 0.0079008563, 0.00046652918, 1.7357044e-05, 3.3269059e-07, 1.7839085e-09
126	2.4, 37.731634, 281.96902, 1332.5318, 4469.3399, 11318.515, 22480.107, 35903.288, 46907.06, 50730.517, 45787.446, 34667.851, 22081.615, 11840.399, 5338.0607, 2016.8262, 635.15342, 165.42827, 35.250985, 6.0565574, 0.82263078, 0.085965852, 0.0066500234, 0.00035927923, 1.2311131e-05, 2.2155984e-07, 1.2368071e-09
128	1, 16.054142, 122.58344, 592.27787, 2032.3121, 5269.0313, 10721.201, 17555.267, 23533.128, 26135.781, 24244.389, 18883.507, 12384.817, 6844.8195, 3184.0114, 1242.6326, 404.72539, 109.16233, 24.124646, 4.306259, 0.60899359, 0.066460691, 0.0053936324, 0.0003081774, 1.1355066e-05, 2.2933981e-07, 1.7006534e-09
130	1.4545455, 23.571819, 181.80481, 887.93976, 3082.3521, 8091.7804, 16688.259, 27727.853, 37764.331, 42673.942, 40345.365, 32090.214, 21542.397, 12220.392, 5854.3133, 2362.744, 799.92111, 225.74379, 52.642355, 10.026535, 1.5359339, 0.18538664, 0.017143406, 0.0011672031, 5.5096348e-05, 1.6327456e-06, 2.5270645e-08, 1.41708e-10

g	$\ell_1, \ell_2, \ldots$
132	2.2222222, 36.917531, 292.27264, 1467.2786, 5243.2814, 14192.156, 30230.137, 51971.535, 73384.65, 86154.384, 84817.351, 70420.443, 49475.774, 29456.253, 14854.997, 6331.7207, 2271.7617, 681.93864, 169.81931, 34.685275, 5.7234643, 0.74779615, 0.075273275, 0.0056170993, 0.00029333105, 9.737256e-06, 1.6905204e-07, 8.2905113e-10
$p_0 = 67$	
134	1.0153846, 16.783934, 132.25269, 661.08587, 2353.3901, 6349.6754, 13492.167, 23160.126, 32688.197, 38410.08, 37906.863, 31609.021, 22353.788, 13431.647, 6857.6644, 2970.1994, 1087.6536, 334.95951, 86.105253, 18.288162, 3.1658034, 0.43856498, 0.047429621, 0.0038683268, 0.00022627152, 8.7782027e-06, 1.9695597e-07, 1.8717451e-09
136	1.0666667, 17.971975, 144.48768, 737.64795, 2684.8056, 7414.549, 16145.161, 28436.239, 41235.398, 49852.878, 50698.471, 43636.067, 31911.189, 19868.86, 10536.331, 4752.7889, 1818.4516, 587.41838, 159.16077, 35.851946, 6.6355079, 0.99354048, 0.11788584, 0.010774575, 0.00072839523, 3.4202316e-05, 9.9806295e-07, 1.404311e-08, 2.5509536e-11
138	2.0952381, 35.530078, 287.51187, 1477.5273, 5413.8581, 15053.727, 33009.445, 58558.798, 85550.333, 104232.8, 106864.04, 92768.371, 68462.535, 43045.706, 23070.172, 10528.247, 4080.4159, 1337.3165, 368.36698, 84.574535, 16.008631, 2.462548, 0.30205268, 0.028794581, 0.0020580578, 0.00010454028, 3.4571879e-06, 6.29087e-08, 4.0600563e-10
140	1.6, 27.854675, 231.68988, 1225.5066, 4628.5667, 13287.137, 30131.906, 55387.106, 84019.037, 106538.77, 113972.78, 103534.18, 80213.934, 53139.352, 30131.114, 14616.456, 6054.3587, 2134.1251, 636.97894, 159.88927, 33.450377, 5.7651907, 0.80633376, 0.089733145, 0.0077389364, 0.00049854302, 2.269507e-05, 6.6405691e-07, 1.0202759e-08, 4.0345487e-11
$p_0 = 71$	
142	1.0144928, 17.542294, 144.93459, 761.49956, 2856.9057, 8146.5815, 18350.849, 33504.663, 50479.124, 63567.834, 67525.925, 60900.499, 46834.35, 30789.134, 17319.182, 8331.4431, 3420.6651, 1194.4755, 352.92575, 87.613718, 18.105255, 3.0768917, 0.42326566, 0.046152457, 0.0038765749, 0.00024078156, 1.0377366e-05, 2.765465e-07, 3.4297029e-09
144	2, 35.144732, 295.28721, 1578.9991, 6034.3255, 17545.112, 40342.761, 75281.179, 116086.07, 149862.53, 163498.41, 151764.18, 120414.06, 81903.47, 47825.217, 23974.948, 10305.024, 3787.8978, 1185.9392, 314.50176, 70.126534, 13.022824, 1.9896725, 0.24618491, 0.02416375, 0.0018295959, 0.00010269868, 4.0174486e-06, 9.7885233e-08, 1.1234938e-09
$p_0 = 73$	
146	1.0140845, 17.933256, 151.72063, 817.43126, 3149.6365, 9239.9782, 21454.372, 40463.185, 63125.721, 82536.924, 91312.227, 86067.688, 69450.324, 48126.71, 28687.727, 14714.719, 6488.7319, 2454.6468, 793.87584, 218.45526, 50.822318, 9.9145474, 1.6050822, 0.21280135, 0.022713552, 0.0019080837, 0.00012225915, 5.7032527e-06, 1.7953059e-07, 3.3072228e-09, 2.5236209e-11
148	1.0285714, 18.044617, 151.43169, 809.20109, 3092.0139, 8994.2022, 20703.52, 38702.68, 59833.493, 77505.926, 84926.13, 79257.312, 63299.953, 43397.866, 25581.883, 12969.375, 5649.4559, 2109.7618, 673.0952, 182.55958, 41.821462, 8.0247534, 1.2760721, 0.16588478, 0.017318338, 0.0014175966, 8.7906792e-05, 3.9129601e-06, 1.1346584e-07, 1.7208572e-09, 5.4649147e-12
150	2.6666667, 48.065624, 414.78795, 2281.3064, 8980.6956, 26941.87, 64031.895, 123740.14, 198018.89, 265898.77, 302501.08, 293617.24, 244361.63, 174947.63, 107947.77, 57438.555, 26339.934, 10391.569, 3516.6253, 1016.564, 249.63469, 51.709748, 8.955709, 1.2826605, 0.14984025, 0.014028248, 0.0010280418, 5.6998117e-05, 2.261566e-06, 5.7652929e-08, 7.1382142e-10

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